

Exoknee Introduction

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Overall System Design

The architecture presented in this work is inspired by biological observations of the human musculoskeletal-joint complex. Accordingly, the proposed knee actuator prioritizes the following functional characteristics:

1. **Biomimetic Elasticity:** Low-cost, commercial polyurethane (PU) cords are employed as compliance elements to mimic biological muscle tissue, providing passive compliance and intrinsic mechanical safety.
2. **Hierarchical Recruitment:** Actuation bundles are recruited sequentially in discrete stages, directly mirroring biological motor unit recruitment patterns.
3. **Discretized Stiffness Modulation:** System stiffness is governed dynamically via selective binary switching (in/out) of active PU elastic cords.
4. **Linearization via Binary Coding:** A binary-weighted arrangement is leveraged to achieve digitally linear dynamic stiffness regulation.

Specific Implementation

The proposed knee exoskeleton comprises three primary sub-assemblies:

1. A lower leg brace (LLB)
2. A 3-DOF hard-link kinematic structure (3DHL)
3. An elastic driver system, comprising:
 - A Brushless DC (BLDC) motor
 - A binary-coded PU elastic cord matrix
 - A digitally controlled cam-brake system

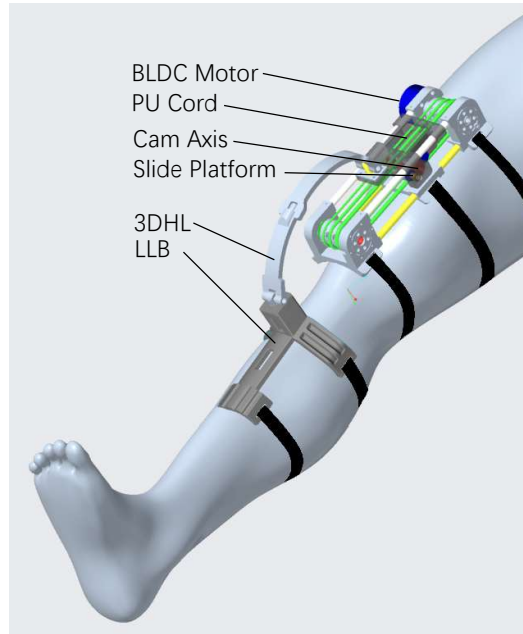


Figure 1 – Overall Design of the Knee Exoskeleton

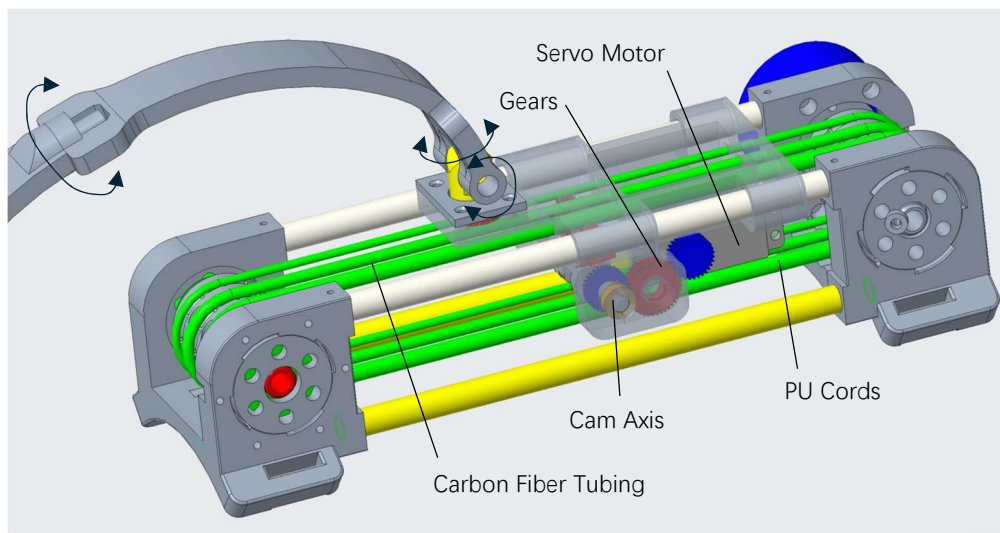


Figure 2 – The Elastic Driver

Detailed Design

Binary PU Cord System

In this research prototype, the PU elastic cord system includes 3 PU cords, with diameters of 3mm, 4.24mm, and 6mm.

Table 1 – Binary Coding with PU Cord Diameter

PU Cord	Diameter (mm)	Area(mm ²)
Cord 1	3	9
Cord 2	4.24	18
Cord 3	6	36

So the cross-section areas of these PU cords are 9, 18, and 36 mm², respectively. The ratio of these

cross-section areas, and therefore of stiffnesses, is then 1:2:4. Because the stiffness is proportion to the area of the cross-section, the ratio of the stiffness is also 1:2:4.

Binary Cam Control

In order to control the switch in and out of these PU cords, a binary CAM is designed to implement binary coding with rotation angles. The CAM axis operates with 8 discrete angles: 0, 45, 90, 135, 180, 225, 270, 315 degrees. On the CAM axis there are 3 cams with high and low height positions. For each angle, the heights of these CAMs form a binary coding as shown below:

Table 2

Angular position	Cam1 height	Cam2 height	Cam 3 height	Stiffness Coefficient
0	low	low	low	0
45	low	low	high	1
90	low	high	low	2
135	low	high	high	3
180	high	low	low	4
225	high	low	high	5
270	high	high	low	6
315	high	high	high	7

When a cam is high, it presses down on a brake on to a PU cord, binding the sliding platform to the PU cord.

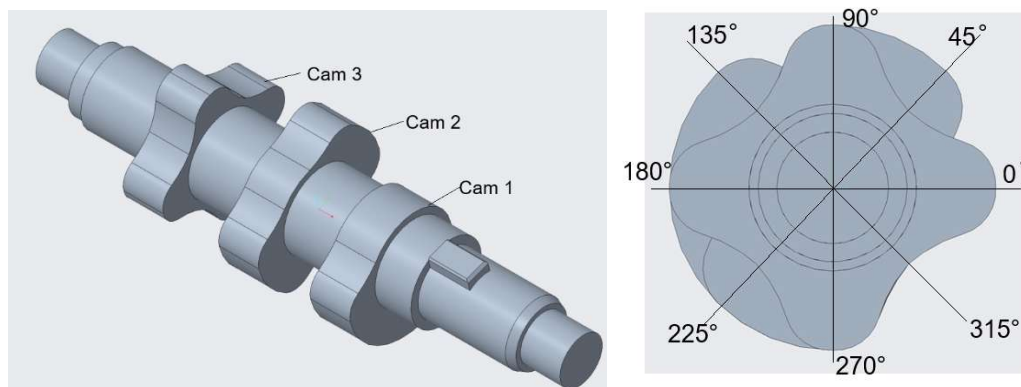


Figure 3 – 8-Angle Cam system